



ETHICAL CONSIDERATIONS OF DATA PRIVACY IN SMARTWATCH TECHNOLOGY

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Introduction

- Smartwatches users continues to grow exponentially each year
- Smartwatch data contributes to both user satisfaction and medical advancement
- Inappropriate use of this data can have dangerous consequences
- Ensuring data privacy is crucial for proper use of smartwatch data
- Tools and techniques can be adopted to safeguard data privacy and security

Smartwatches: The Current State of Play

Market Size

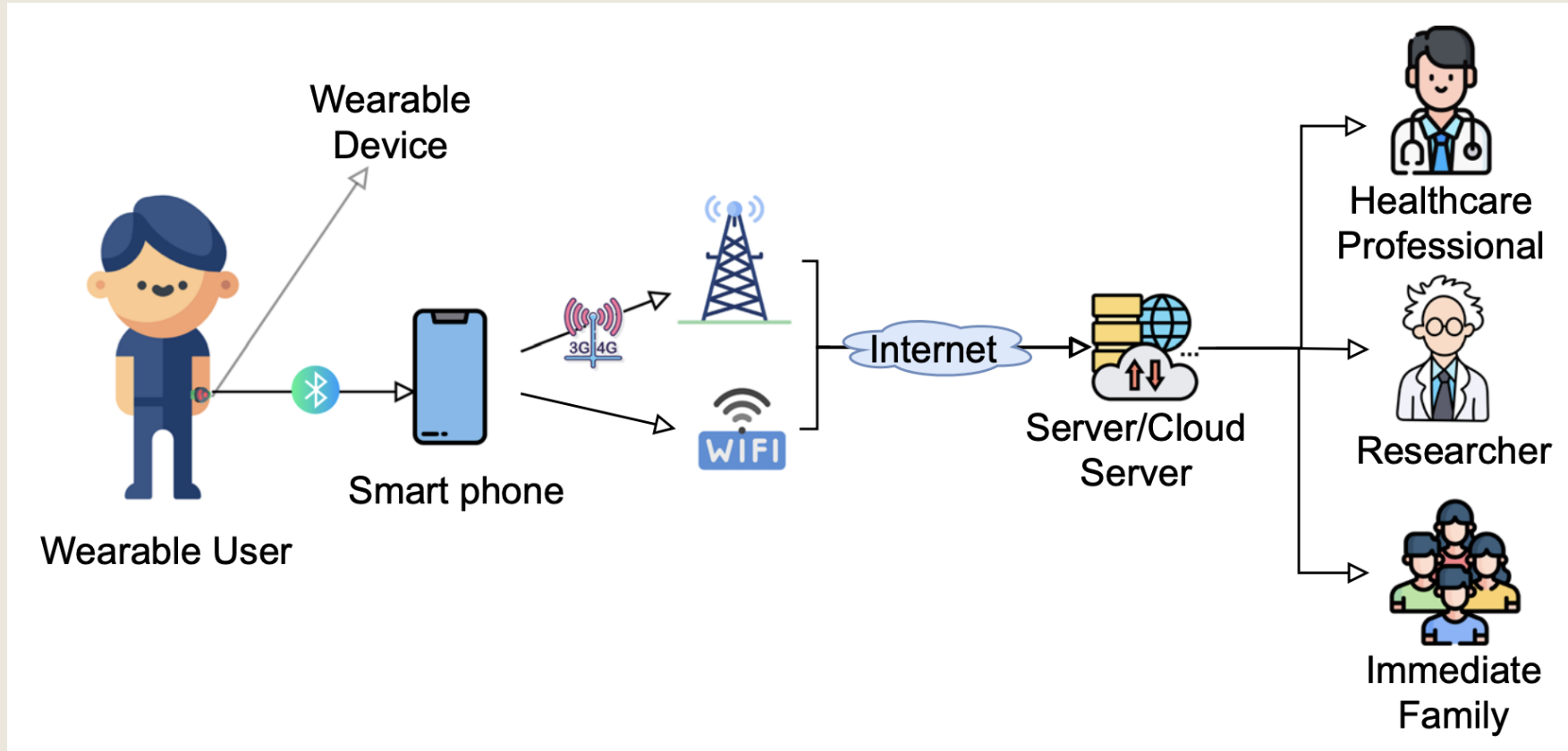
- Over 500 million wearable devices in use worldwide as of 2025
- Smartwatches make up 44% of wearable device market
- 740 million global smartwatch users projected by 2029

Smartwatches: The Current State of Play

Big Players

- Fitbit first developed in 2009
- Samsung's first smartwatch released in 2013
- Google acquired Fitbit in 2019, later launching its first smartwatch in 2022
- Apple Watch developed in 2015 and remains the most popular manufacturer

Wearable Device Architecture



What benefits do smartwatch data provide?

- Personal activity and sleep tracking
- Disease detection
- Medical research
- Public health initiatives
- Early detection of pandemics
- Mental and physiological benefits for users

What privacy issues and risks exist with smartwatch data?

- Insecure collection and storage of person-generated health data (PGHD)
- Unregulated nature of PGHD both in the U.S. and abroad
- Intentionally long and confusing privacy policies deceiving users

What encompasses person-generated health data (PGHD)?

- Sensor-based data
 - *Sleep patterns*
 - *Activity levels*
 - *Heart rate*
- User-reported data
 - *Diet*
 - *Stress levels*
 - *Wellbeing*
- Socioeconomic factors with indirect relevance to health

Smartwatch PGHD collection and storage is insecure

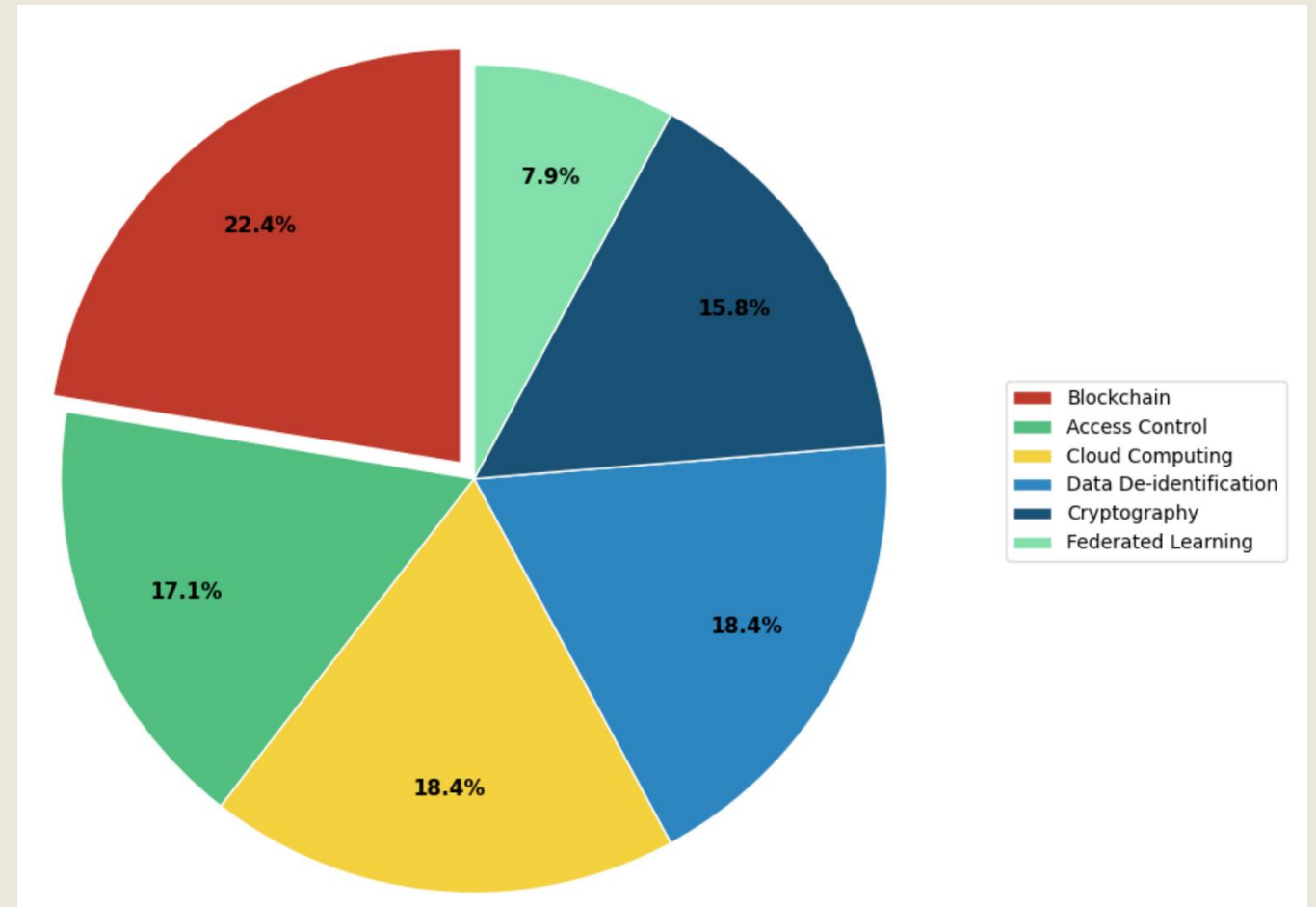
- 44% of all data breaches from 2005-2019 came from the healthcare industry
- Wearable data is particularly vulnerable to hackers and breaches in these stages
 - 2018: *Strava military hack*
 - 2021: *GetHealth data leak*
 - 2022: *Google privacy breach*
- Over 170 million U.S. data breaches in 2023

Why is PGHD from smartwatches not secure?

- Unauthorized data collection
- Flawed access control systems
- Improper anonymization techniques
- Existence of financial and private information in smartphones

Privacy Enhancing Technologies

- **Blockchain** decentralizes data management
- **Federated Learning** leverages data localization
- Data de-identification frameworks like **Differential Privacy** utilize random noise to ensure anonymization



Frequency of privacy protection technologies in 52 research papers

Limitations of Privacy Enhancing Technologies

- Blockchain ensures data records remain unchanged, **but requires extra computing power that may impact application performance**
- Differential Privacy can essentially guarantee anonymity, **but with a tradeoff of data utility loss**
- Federated Learning preserves data locality, **but it is vulnerable to data leakage during training**

PGHD from smartwatches is unregulated

- Minimal HIPAA regulations
- Lack of FDA oversight
- Regional disparities, both nationally and globally
- Obscure third-party data practices

Current State of Wearable Device Regulations

What's In Place

- HIPAA applicable if the wearable device is prescribed by a doctor
- FDA oversight only required for medical devices
- Regional and state-level regulations

What's Not Covered

- Data from voluntary smartwatch usage not protected under HIPAA
- No FDA oversight for smartwatch data
- No federal/global regulations

Why are PGHD regulations important?

- For companies to fend off cyber attacks that compromise their enterprise data
- For smartwatch users to be safe from having their identities stolen
- To protect against misguided use of PGHD and resale to third parties
- To avoid discriminatory use of PGHD by insurance companies and employers
- To protect vulnerable populations

Case study of privacy risk from unregulated PGHD

In 2024, a data brokerage company called Near Intelligence sold geo-location data of individuals' Planned Parenthood visits to an anti-abortion group.

The group then used this data to organize anti-abortion campaigns targeted at these individuals.

Privacy policies are intentionally long and complex

- Up to 97% of people agree to terms and conditions without fully understanding them
- If asked to share personal data during app installation, 92% of users will provide
- A 2025 study estimates it would take 26 minutes for a user to read the average wearable device privacy policy

Privacy Paradox

The phenomenon in which users claim to care about privacy but put their data at risk in order to gain access to products and services

- Lack of policy comprehension
- Unfamiliar devices/technologies
- Convenience mindset

Effects of Privacy Policy Confusion

- 82% of people in the U.S. are unaware that apps can sell data users' PGHD
- Policy complexity leads to privacy fatigue and a lack of meaningful consent
- Privacy paradox lets businesses believe that users value access over privacy
- Users are unaware of potential third-party purchase and dissemination of their data
- 67% of users in a 2025 study were uninformed that sensitive information can be shared from smartphone to smartwatch

What can be done to prioritize smartwatch data privacy?

- Increased federal and industry regulations
- Requirement of privacy enhancing technologies
- Creation of a standardized privacy policy certification system
- Transparency reporting

Federal and Industry Regulations

- Congress and data brokers should work in tandem to create a comprehensive federal privacy framework
 - *General Data Protection Regulation (EU)*
 - *California Consumer Privacy Act*
- The Federal Trade Commission should be leveraged to supervise privacy execution and reprimand unethical/exploitative practices
- Industry regulations/best practices developed and implemented by a respected and relevant non-government organization

Privacy Enhancing Technology Requirements

- According to recent reviews, more health companies are adopting data anonymization techniques of their own accord
- Still, within the federal and institutional regulations, some form of data anonymization should be a required practice for data collection and storage
 - *Blockchain* – to shift control of PGHD back to users
 - *Federated Learning* – to allow collaboration across institutions without jeopardizing data ownership
 - *Local Differential Privacy* – to anonymize data prior to arriving at central server

Standardized Privacy Policies

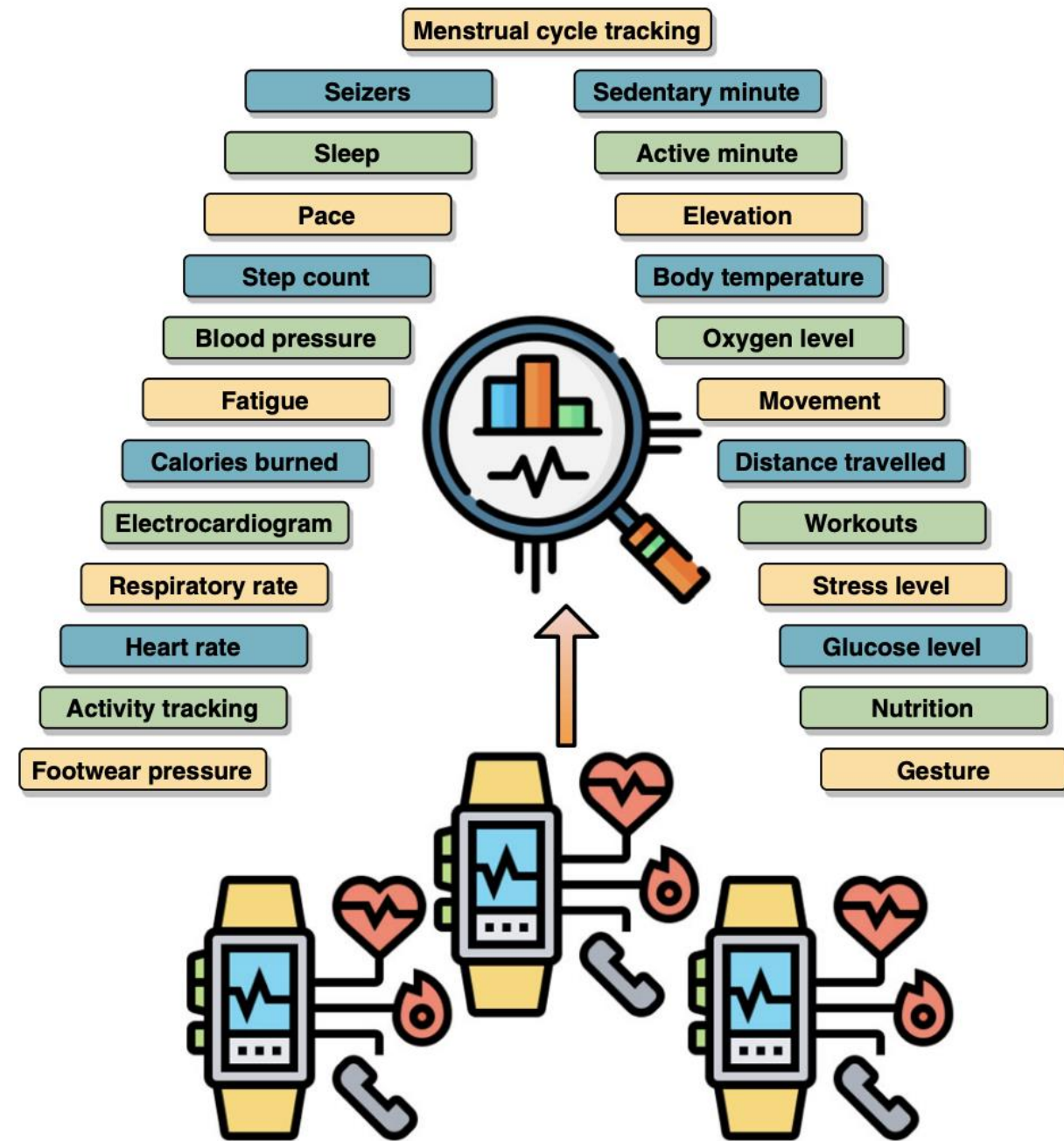
- Privacy policies should increase adoption of consumer-friendly language
- Maximum user privacy protection should be the default setting for smartwatches
 - *Broad and uninformed consent is not meaningful, which undermines autonomy and legitimacy*
- A tier-system should be developed to label privacy policies
 - *‘Bronze’, ‘silver’, and ‘gold’ certifications*
 - *‘Maximal Privacy’, ‘Balanced’, and ‘Performance Optimized’*

Transparency Reporting

- Requirements should be implemented for smartwatch companies to provide documentation related to privacy practices
 - *How data is being used*
 - *Who has access to the data & why is this access legally justified*
 - *Rates of compliance*
- Reporting should be ‘living’ assessments, tracking company actions, not words

Conclusion

- Smartwatches collect previously unseen levels of personal health data
- This data is important to consumers as well as to medical research
- Ensuring society reaps all the benefits of this data while protecting consumer data privacy is of the utmost importance



Conclusion

- As smartwatch use continues to expand, ethical considerations need to expand
- Increased regulations will allow for standardized privacy practices
- Data anonymization is crucial for protecting consumer privacy
- Privacy policies must prioritize explicit, meaningful consent
- Companies must be transparent and forthcoming in their reporting

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